EDGE QUANTUM GUARD (EQG)
Technical Validation Report

Hardware validation of the full four-role control stack
on IBM Quantum (**ibm_fez**)

Project	Edge Quantum Guard (EQG)
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Workload	QAOA MaxCut — path graph, N = 8, p = 2
Backend	ibm_fez (IBM Quantum, real QPU)
Experiments	R1: B→A (7 s/arm) · R2: A→B (15 s/arm)
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Abstract. Two controlled hardware runs of the same QAOA workload — with and without the full EDGE Quantum Guard stack — yield statistically significant improvements in approximation ratio (z = 5.7 and z = 27.7) and measurable gains in layout stability. The inverted arm order (A→B) reproduces the positive effect, reducing the temporal-drift confound. This report documents design, results, attribution, and product-relevant limits.

1. Objective

Demonstrate, on real quantum hardware, that the **complete EQG mechanism** (four roles) measurably improves (i) solution quality of a useful workload (QAOA approximation ratio) and (ii) effective layout stability/coherence during and after execution, versus the **same workload** without EQG (naive layout, no dynamical decoupling, no inter-job reaction).

The goal is preliminary causal evidence for a low-level control protocol on NISQ hardware — not universal generalization across all devices or algorithms.

2. System under test

EQG is structured as an autonomous low-level control stack:

Role	Module	Function in these runs
Eyes	AOM / health	Friction, regime, per-qubit health (calibration + measurement)
Reflexes	MORPH	Best connected physical path (qubit health + CZ error)
Judgment	Minimal plan + causal	Conditional DD, remap, or hold; causal verdict
Hands	DD CPMG/XY4 + exec.	Idle-window DD insertion and QAOA execution

Pre-registered significance threshold: $\mathbf{z \geq 2}$ on $\Delta AR = AR(B) - AR(A)$.

3. Experimental design

3.1 Common protocol

`PRE (simultaneous snapshot) → Arm 1 → MID → Arm 2 → POST`

Identical QAOA workload and **identical shot plan** on both arms. PRE/MID/POST snapshots measure simultaneously: MORPH layout, MORPH+DD, and naive layout. Effect separation: **layout** (B – A at the same instant) vs **DD** (B+DD – B).

3.2 Independent runs

Run	Order	Budget / arm	QAOA shots / arm (approx.)	Calibration
R1 — reference	B → A	7.0 s	~2,800	2026-10-10
R2 — inverted replica	A → B	15.0 s	~25,200	2026-10-12

Arm-order inversion controls the confound of chip temporal drift between sequential arms.

4. Results

4.1 Solution quality (QAOA)

Metric	R1 (B→A, 7 s)	R2 (A→B, 15 s)
AR(B) EQG ON	0.8147 ± 0.0027	0.8092 ± 0.0009
AR(A) EQG OFF	0.7923 ± 0.0029	0.7713 ± 0.0010
ΔAR (B–A)	+0.0224 ± 0.0040	+0.0379 ± 0.0014
z-score	5.7	27.7
Verdict	EQG_MEJORA_QAOA	EQG_MEJORA_QAOA
Retention B / A	0.95 / 0.93	0.95 / 0.90
P(optimal) B / A	0.221 / 0.185	0.212 / 0.156

In **both** runs ΔAR > 0 with z ■ 2. The improvement is not explained by shot noise.

4.2 Stability during the run

Metric	R1	R2
AR drift arm B	+0.0036 (stable)	+0.0096 (stable, rising)
AR drift arm A	–0.0013 (stable)	–0.0050 (stable, falling)
Instance s mean B / A	0.716 / 0.656	0.744 / 0.645

Under EQG, arm B not only achieves higher mean AR; in R2 it **improves over time**, while the baseline drifts slightly downward.

4.3 Attribution (simultaneous snapshots)

Effect on stability s	R1	R2
Layout MORPH vs naive	+0.033 (significant)	+0.104 (significant)
DD on vs off	+0.061 (significant)	+0.073 (significant)

Both stack components contribute measurably. Layout typically dominates solution-quality gain; DD reinforces effective coherence of the protected layout.

4.4 Judgment behavior (R2, A→B)

- B job 0: stable regime → **no intervention** (DD off).
- Between jobs: coherence/health signal → **conditional DD** (28 pulses in job 1).
- AR on B: 0.805 → **0.815** after enabling DD.
- Causal verdict: `exito=True` (intervention consistent with the signal).

This demonstrates a **reactive** mechanism, not only a static selector at t = 0.

4.5 Acceptance criteria

In R1 and R2: **C1–C7 = OK** (full stack, budgets, AR/stability series, PRE/MID/POST snapshots, statistical verdict).

5. What these tests demonstrate

5.1 Demonstrated (within experimental scope)

1. **Reproducible AR improvement** for the same QAOA workload when full EQG operates versus baseline without EQG, on ibm_fez.
2. **Robustness to arm order:** effect sign holds for B→A and A→B; not explained solely by EQG running "first" in the reference.
3. **Partial mechanism attribution:** MORPH layout and DD contribute detectably to stability metric s.
4. **Reactive control:** Judgment can hold and later enable DD based on measurement, with observable AR impact.
5. **Budget and formal criteria** compliance under pre-registered thresholds.

5.2 Not demonstrated (explicit limits)

1. Universal causality or full independence from chip drift (sequential arms; MID mitigates, does not remove).
2. Generalization to other backends, problem sizes, algorithms (e.g. QSVM), or calibration schedules.
3. Optimality of thresholds or DD sequence versus alternatives.
4. Industrial scale (queue latency, multi-tenant SLOs, continuous product operation).

A pair of runs is **strong evidence of mechanism viability**, not a closed product certification.

6. Product-viability reading

Product question	Reading from R1 + R2
Measurable value vs running the job alone?	Yes, on this workload/backend: ΔAR on the order of 2–4 points with high z.
Only a fixed layout at t = 0?	No — inter-job reaction and conditional DD with observed effect.
Survives temporal order confound?	Yes — inverted A→B replica preserves positive sign.
Ready for broad commercial claim?	Not yet — need replica series, formal ablations, ≥1 more backend or workload.

Recommended position. EQG as a **low-level stability/coherence control layer** with positive experimental evidence on real NISQ hardware — suitable for a technical report, pilots, and the next validation phase (replicas, MORPH-only / DD-only / full-stack ablation, additional crossed order).

7. Conclusion

Runs R1 (B→A, 7 s) and R2 (A→B, 15 s) on **ibm_fez**, with full EQG and the same QAOA workload, consistently produce the verdict **`EQG_MEJORA_QAOA`**, with statistically significant approximation-ratio gains and positive attribution of layout and DD on stability metrics.

These results **demonstrate experimental viability of the EQG mechanism** as a four-role system to improve useful execution and effective coherence on real hardware, under the stated design limits. They do not replace a multi-replica validation program for a general product claim, but they **do form the quantitative basis** to support EQG as a technology demonstrated under controlled QPU conditions.

8. Artifacts and job identifiers

R1 outputs: eqg_qaoa/report.json, summary.txt, qaoa_stability.png

R2 outputs: eqg_qaoa_AB/report.json, summary.txt, qaoa_stability.png

R2 job IDs (ibm_fez):

PRE db6557o4qg6s73c489eg · A db655aamb58s738b8vdg, db655dkvf2bc73d0m370 · MID db655hklf4us73c4qfpg · B db655k4lf4us73c4qft0, db655nclf4us73c4qg00 · POST db655rclf4us73c4qg4g

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